

Network Intrusion Detection on the CICIDS2017 Dataset Using Random Forest Classification

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Abstract:

This project explores the application of machine learning to network intrusion detection using the CICIDS2017 dataset which is a widely used benchmark in cybersecurity research. A Random Forest classifier was trained and evaluated on network flow data to distinguish between benign and malicious traffic, first as a binary classification task (Normal vs. Attack) and then as a multiclass task across seven traffic categories (Normal Traffic, DoS, DDoS, Port Scanning, Brute Force, Web Attacks, and Bots). While the model achieved near-perfect performance on aggregate metrics, deeper investigation revealed that this result is one of the documented limitations of the CICIDS2017 dataset. A deeper class-level analysis uncovered a specific and meaningful weakness in detecting Bot traffic, which was frequently misclassified as benign. Feature importance analysis further showed that the model relies heavily on packet-length statistics, and this pattern was consistent across both binary and multiclass models. This report documents the dataset, methodology, results, and interpretation of these findings, along with their implications for real-world applicability.

Objectives:

- Train a machine learning model capable of distinguishing between benign and malicious network traffic using the CICIDS2017 dataset.
- Extend the model to classify traffic into specific attack categories rather than a simple binary label.
- Critically evaluate model performance beyond surface-level accuracy, using precision, recall, F1-score, and confusion matrices.
- Investigate the underlying causes of unusually high-performance scores.
- Identify which network traffic features most influence the model's predictions and assess whether these features reflect genuine attack behaviour or artifacts of the dataset's generation process.
- Draw honest, evidence-based conclusions about the model's strengths, weaknesses, and limitations regarding real-world generalizability.

Dataset:

This project uses the **CICIDS2017** dataset. The version of the dataset used in this project was a pre-processed CSV file sourced from Kaggle, containing 2,520,751 labeled network flow records across 53 columns. Each row represents a single network flow, with features extracted using CICFlowMeter (e.g., packet length statistics, flow duration, inter-arrival times, and flag counts) and a label indicating the traffic type: Normal Traffic, DoS, DDoS, Port Scanning, Brute Force, Web Attacks, or Bots. [Download used Dataset here.](#)

Procedure:

1. Data Loading and Exploration:

The dataset was loaded using pandas, and column names were cleaned to remove extraneous whitespace. The label column's class distribution was examined to understand the composition of the dataset before any modelling was attempted. This step revealed a substantial class imbalance: normal traffic accounted for approximately 83% of all records, while some attack categories, notably Bots and Web Attacks, made up less than 0.1% each.

Label Distribution:

Attack Type	count
Normal Traffic	2095057
DoS	193745
DDoS	128014
Port Scanning	90694
Brute Force	9150
Web Attacks	2143
Bots	1948

The dataset was also checked for missing and infinite values, both of which are common issues with this dataset due to how certain flow features (e.g., bytes/second) are calculated. No such issues were found in this particular pre-processed version.

2. Data Preparation:

Features (X) were separated from the target label (y) to prevent the model from having direct access to the answer during training. The label was first simplified into a binary format (Normal Traffic vs. ATTACK) to establish a baseline before attempting the more complex multiclass problem.

The dataset was split into training (70%) and test (30%) sets using stratified sampling, which preserves the original class ratio in both subsets. This was particularly important given the scale of class imbalance present, as a non-stratified split risked underrepresenting rare attack types in the test set.

3. Model Training:

A Random Forest classifier (100 estimators) was selected for this task due to its strong baseline performance on tabular data, minimal preprocessing requirements, and built-in support for feature importance analysis. The model was trained separately for the binary and multiclass classification tasks.

4. Evaluation:

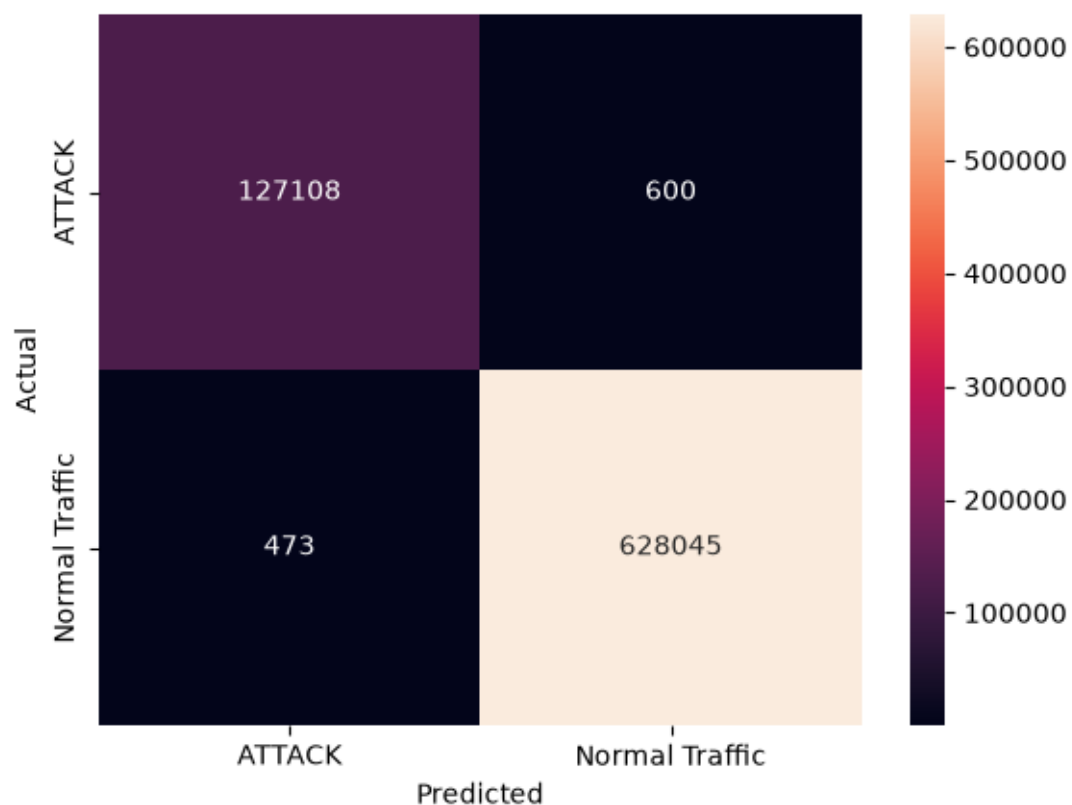
Model performance was evaluated using precision, recall, and F1-score, since accuracy is a misleading metric under significant class imbalance. Confusion matrices were also generated to examine the exact distribution of correct and incorrect predictions at the individual class level.

a) Binary classification results:

Binary classification results showed precision, recall, and F1-scores that rounded to 1.00 for both classes. Given how unusual such a result is in applied machine learning, this was treated as a finding to investigate rather than a result to accept outright.

	precision	recall	f1-score	support
ATTACK	1.00	1.00	1.00	127708
Normal Traffic	1.00	1.00	1.00	628518
accuracy			1.00	756226
macro avg	1.00	1.00	1.00	756226
weighted avg	1.00	1.00	1.00	756226

Confusion matrix:



Two possible explanations were tested directly:

- **Duplicate records:** Only 161 duplicate rows were found out of over 2.5 million, ruling this out as a meaningful cause.
- **Metadata leakage via Destination Port:** Because CICIDS2017 attacks were often directed at fixed, specific ports within the testbed, this feature was suspected as a possible shortcut. Removing it entirely and retraining the model produced no change in performance, ruling this out as well.

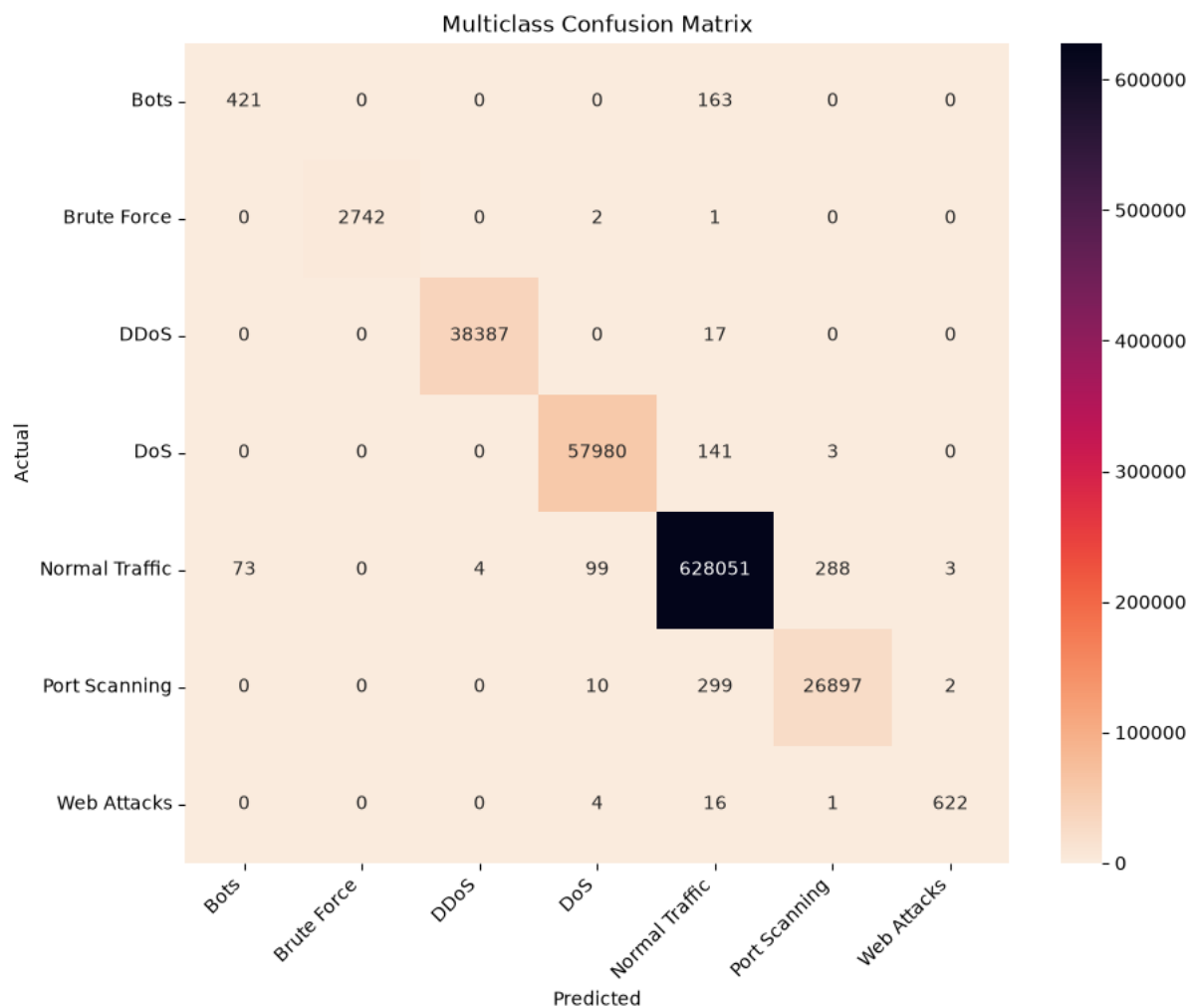
With both explanations ruled out, literature on CICIDS2017 was reviewed, which confirmed that near-perfect scores are a widely reported pattern with this dataset (see References). This is generally attributed to the dataset's attacks being generated using automated tools in a controlled environment, producing highly consistent statistical signatures that are easier to separate than real-world, more varied attack traffic would be.

b) Multiclass classification results:

Multiclass classification results provided a more detailed and informative picture. While most classes (DoS, DDoS, Brute Force, Port Scanning) were classified with near-perfect precision and recall, the Bots category stood out with a recall of only 0.72, notably weaker than every other class, including some smaller classes. This indicated that the weakness was not simply a function of limited training data, but something more specific to how Bot traffic behaves.

	precision	recall	f1-score	support
Bots	0.85	0.72	0.78	584
Brute Force	1.00	1.00	1.00	2745
DDoS	1.00	1.00	1.00	38404
DoS	1.00	1.00	1.00	58124
Normal Traffic	1.00	1.00	1.00	628518
Port Scanning	0.99	0.99	0.99	27208
Web Attacks	0.99	0.97	0.98	643
accuracy			1.00	756226
macro avg	0.98	0.95	0.96	756226
weighted avg	1.00	1.00	1.00	756226

Confusion Matrix:

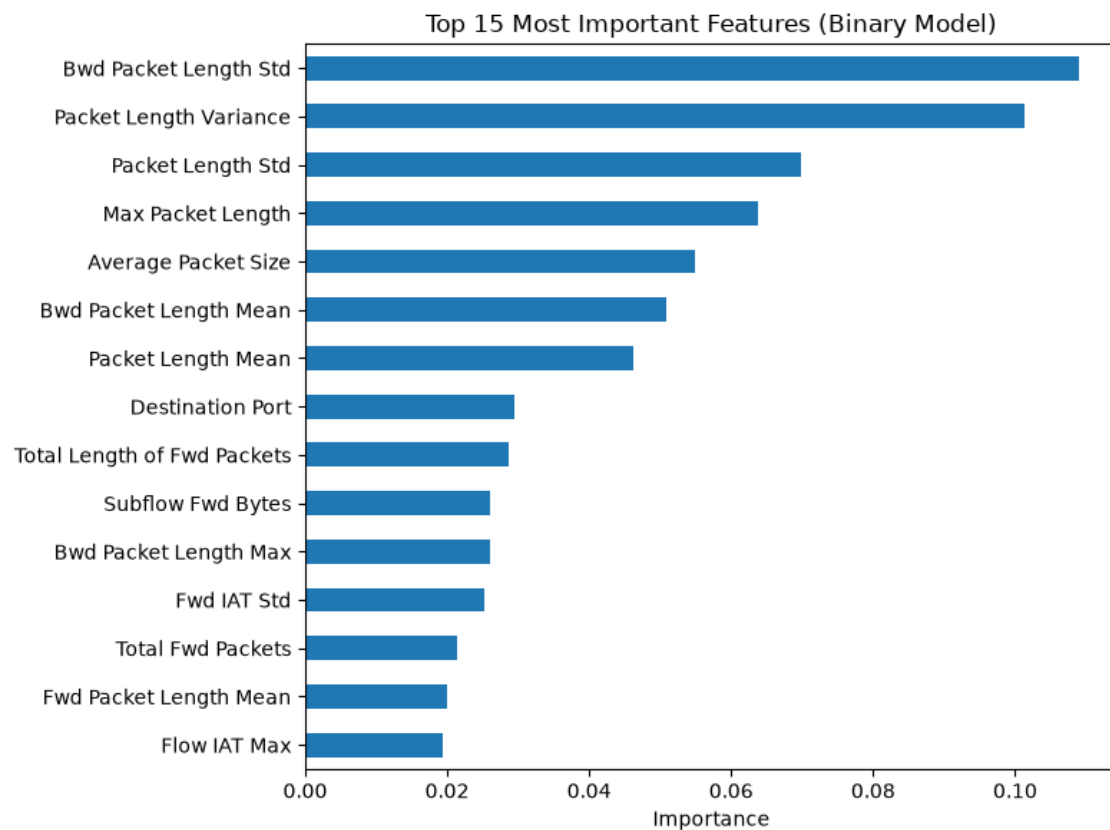


A closer look at the multiclass confusion matrix showed that nearly all misclassified Bot samples were predicted as Normal Traffic, and not confused with any other attack category. This suggests that Bot traffic in this dataset closely resembles benign traffic from a statistical standpoint, which is consistent with the behavior of real botnets that are often designed to blend in with normal network activity to avoid detection.

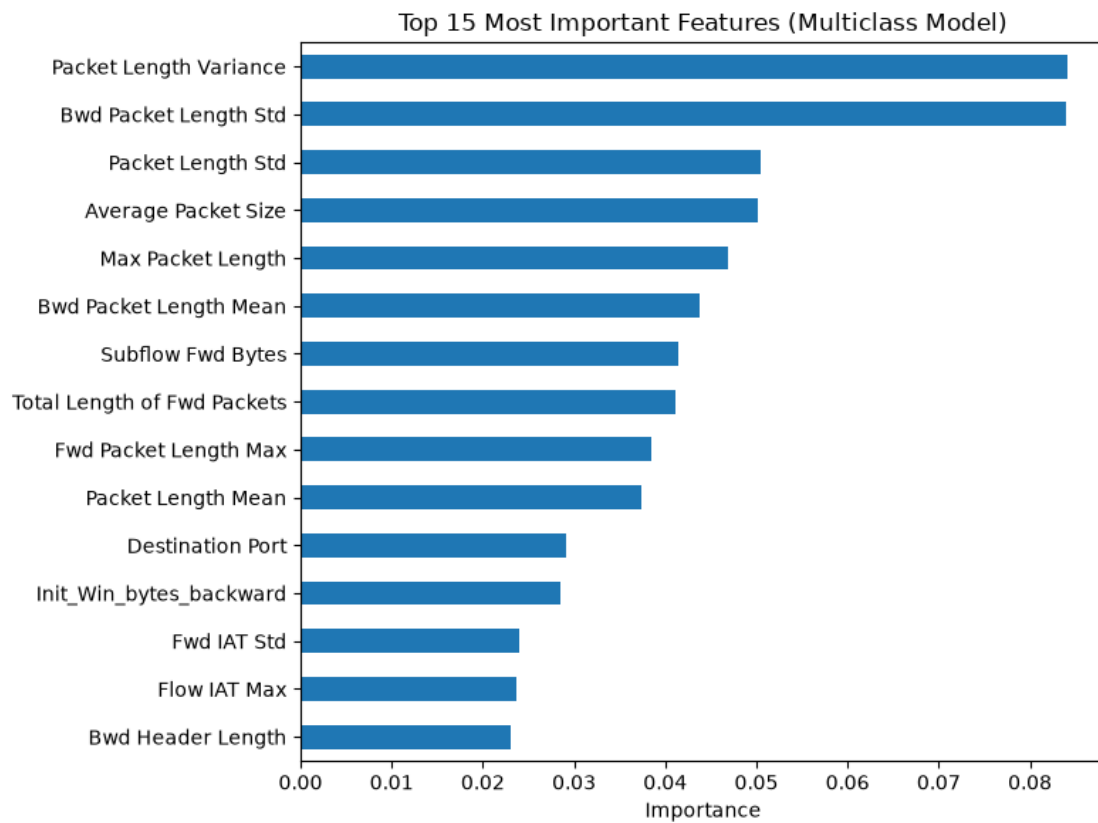
5. Feature Importance Analysis:

To understand what the model actually relies on to make predictions, feature importance scores were extracted for both the binary and multiclass models. In both cases, packet-length-related statistics, including packet length variance, standard deviation, and average packet size, accounted for a substantial share of the model's decision-making, together representing roughly a third of total feature importance. Destination Port ranked comparatively low in both models, further supporting the earlier finding that it was not a significant driver of performance.

Importance bar chart for binary model:



Importance bar chart for multiclass model:



This finding aligns with concerns raised in the literature: because the attacks in CICIDS2017 were generated using specific automated tools, they tend to produce distinct and consistent packet-size patterns that may not align with more varied, real-world attack traffic.

6. Key Findings:

- The model achieves near-perfect aggregate performance, but this reflects a documented limitation of the CICIDS2017 dataset rather than a fully solved intrusion detection problem.
- Despite high aggregate scores, class-level analysis revealed a genuine weakness in detecting Bot traffic (recall of 0.72), which was consistently confused with normal, benign traffic rather than other attack types.
- Class size alone does not fully explain detection difficulty, some smaller classes (e.g., Brute Force) were classified perfectly, while Bots, a comparably sized class, performed considerably worse. This points to differences in the underlying statistical separability of traffic types, not just the amount of training data available.
- The model's decisions are driven primarily by packet-length statistics rather than metadata such as Destination Port, a pattern consistent across both binary and multiclass models.
- These findings underscore the importance of evaluating machine learning models on security datasets with skepticism, using controlled experiments and literature context rather than accepting high performance metrics at face value.

7. Limitations and Future Work:

The primary limitation of this project is that it evaluates the model using a random train/test split, which allows flows from the same attack session to appear in both the training and test sets. A stronger evaluation would use a temporal, day-based split, training on earlier days of traffic and testing on a later, unseen day, which better simulates how a model would perform on genuinely new network activity. Future work could also explore unsupervised anomaly detection methods, which do not rely on pre-labeled attack data and may better reflect real-world detection scenarios where novel attack types are not previously known.

8. References:

Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization. *4th International Conference on Information Systems Security and Privacy (ICISSP)*, Portugal, 108–116.

Lanvin, M., Gimenez, P. F., Han, Y., Majoreczyk, F., Mé, L., & Totel, É. (2023). Errors in the CICIDS2017 Dataset and the Significant Differences in Detection Performances It Makes. In *Risks and Security of Internet and Systems (CRiSIS 2022)*, Lecture Notes in Computer Science, vol. 13857. Springer, Cham.